



|    |        |     |       |    |       |  |  |  |  |  |
|----|--------|-----|-------|----|-------|--|--|--|--|--|
| 2  | linear | 20  | scale | 4  | 0.57  |  |  |  |  |  |
| 3  | linear | 30  | scale | 5  | 0.6   |  |  |  |  |  |
| 4  | rbf    | 100 | scale | 5  | 0.35  |  |  |  |  |  |
| 5  | rbf    | 10  | auto  | 10 | -0.04 |  |  |  |  |  |
| 6  | rbf    | 20  | auto  | 5  | 0.01  |  |  |  |  |  |
| 7  | poly   | 20  | scale | 5  | 0.25  |  |  |  |  |  |
| 8  | poly   | 50  | scale | 3  | 0.46  |  |  |  |  |  |
| 9  | poly   | 50  | scale | 1  | 0.5   |  |  |  |  |  |
| 10 | poly   | 60  | scale | 10 | 0.63  |  |  |  |  |  |

The Support Vector Machine uses  $R^2$  value (with hyperparameters kernel=poly, C=60, gamma=scale, degree=10) = 0.63

#### 4.DECISION TREE:

| <u>R.No</u> | <b>Criterion</b> | <b>Splitter</b> | <b>Max Depth</b> | <b>R Score</b> |  |  |  |  |  |  |
|-------------|------------------|-----------------|------------------|----------------|--|--|--|--|--|--|
| 1           | friedman_mse     | best            | 3                | 0.87           |  |  |  |  |  |  |
| 2           | friedman_mse     | best            | 5                | 0.82           |  |  |  |  |  |  |
| 3           | friedman_mse     | best            | 10               | 0.73           |  |  |  |  |  |  |
| 4           | squared_error    | best            | 6                | 0.81           |  |  |  |  |  |  |
| 5           | squared_error    | best            | 8                | 0.74           |  |  |  |  |  |  |
| 6           | squared_error    | best            | 20               | 0.69           |  |  |  |  |  |  |
| 7           | absolute_error   | best            | 5                | 0.85           |  |  |  |  |  |  |
| 8           | absolute_error   | best            | 2                | 0.85           |  |  |  |  |  |  |
| 9           | absolute_error   | best            | 1                | 0.67           |  |  |  |  |  |  |
| 10          | poisson          | best            | 4                | 0.58           |  |  |  |  |  |  |
| 11          | poisson          | best            | 3                | 0.87           |  |  |  |  |  |  |

The Decision Tree Regression uses  $R^2$  value (with hyperparameters criterion=friedman\_mse, splitter=best, max\_depth=3) = 0.87

#### CONCLUSION:

For this dataset, Decision Tree Regression and Random Forest Regression achieved the highest  $R^2$  score of 0.87.